Shiv Nadar University, Chennai

Department of Computer Science

25CS2741 Fundamentals of Internet of Things + Lab

Date: 27/07/2026 (Week 2) **Continuous Lab Evaluation-3** II Yr/ 3rd Sem B. Tech CSE(IoT)A&B

Topics: 8051 Tool Chain setup – Development Board – AT89S52

Strictly No to using **Arduino APIs, HAL libraries, or RTOS services!**

Supporting Documents: **1.** Board Schematics 2. Pin diagram of 8051

Hardware Debug: Multi meter.

Timer-Based LED Pattern Generator and Button Interrupt Controller

Activity:

Design and implement a register-level GPIO controller for a microcontroller development board. Configure four LEDs as digital outputs and a push-button as a digital input. Using only bare-metal Embedded C and direct memory-mapped register access, develop a four-state LED control system. Each valid button press must advance the system to the next operating mode: running LED pattern, binary counter, alternating pattern, and all-LED flashing. Implement switch debouncing and ensure that one physical button press produces exactly one state transition.

Problem Statement

You are developing the first user-interface subsystem for an IoT sensor node.

The system contains:

- **Four LEDs** connected to four GPIO pins.
- **One push-button** connected to a GPIO input.
- The microcontroller must respond to button presses by changing the LED operating mode.

Required Behaviour:

Mode 1 — Running LED	**Mode 2 — Binary Counter**	**Mode 3 — Alternating Pattern**	**Mode 4 — All LED Flash**
Initially, all LEDs are OFF. When the push-button is pressed, the system enters Mode 1: LED1 → LED2 → LED3 → LED4 → LED1 → ...	The four LEDs display a binary count: 0000 0001 0010 0011 ... 1111 0000	On the third button press: 1010 0101 1010 0101	On the fourth button press: 1111 0000 1111 0000

Quick Hacks

1. Check with switch event. Button Pressed?
2. Four Modes as four states.

Deliverable form: Git repo